CLAVE

Euclidean drum machine. 8 touch pads, one encoder and 3 knobs.

User manual · firmware 3.4 · 2026
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> English version of `MANUAL_CLAVE.md`, updated for firmware **3.4** (2026-09-18). Sections marked **new in 3.4** describe what changed since the public 3.0.

Quick start

- Press **play**: the pattern starts.
- Turn up the **master knob** (overall volume).
- Touch a **pad** (1–8) to choose the channel you want to edit.
- Turn the **encoder** to change the number of hits (K).
- **pitch knob** = tuning · **level knob** = volume of that channel.

The 8 channels

1	Kick
2	Snare
3	Closed hi-hat
4	Low tom
5	Conga
6	Cowbell
7	Maracas / shaker
8	Clap

Each pad selects its channel and its LED shows the channel colour. The pad of the channel you are editing stays lit, at half brightness if that channel is muted. The pitch knob has a centre zone that snaps to the sample's original tuning; its LED tells you when you are there. Sequence length is global (1–16, 16 by default).

Controls

Transport and global

play	Play / stop
tempo (tap)	Tap the tempo in quarter notes
master knob	Overall volume

Pattern of the active channel

encoder	Hits K (or the cursor, in linear mode)
swEnc + encoder	Rotate the pattern
swEnc 2 s + encoder	Sequence length (1–16)

Sound of the active channel, and selection

pitch knob	Tuning
level knob	Volume (velocity)
pad 1–8	Choose channel
preset	Load (short) / save (2 s)
shift + preset	Kit selection mode

Secondary functions

Hold a modifier and move the control.

shift +

encoder	Accents (how many)
swEnc + encoder	Rotate accents
swEnc (press)	**Accent the step under the cursor** (linear mode only)
pitch knob	Accent amount (see "Accents" below)
level knob	Pan
mute	Linear / euclidean
tempo	Double tempo (2x)
preset	Kit selection
play short / 2 s	Record / clear automation

tempo +

encoder	BPM (fine adjust)
pitch knob	Channel probability

mute +

pad	Mute that channel
encoder	Effect amount (wet)
swEnc + encoder	Effect mode (14)
pitch / level knob	LP filter: cutoff / resonance

Pattern: probability, length and linear mode

Probability (tempo + pitch knob): 16 = every hit plays · 0 = none · in between = random hit by hit, changing every lap without altering the euclidean pattern.

Sequence length (hold swEnc 2 s): the ring turns amber; turn to choose 1–16 and press swEnc to exit.

Linear mode (shift + mute): place the steps by hand. The encoder moves the cursor and swEnc turns the step under the cursor on or off. It works per channel, so you can have some channels euclidean and others linear in the same song.

Accents in linear mode. In euclidean mode accents spread themselves out, like the hits. In linear mode you draw them: hold shift and press the encoder on the step you want to accent. If that step was off it turns on – an accent without a hit makes no sound.

Changing mode erases nothing. Going from euclidean to linear you inherit the hits and accents exactly as they were playing. Going back to euclidean, the channel keeps playing what you drew: the formula only takes over again when you ask it to spread things out, that is when you add or remove hits, rotate, or change the number of accents.

Accents — new in 3.4

shift + pitch knob sets the accent amount (0–12) for the active channel, and what it does depends on the channel:

Channels	What an accent does	The knob controls
1 kick · 2 snare · 3 hi-hat · 7 shaker · 8 clap	Accented hits play at full volume	How much the **non-accented** hits drop: from –3 dB (left) to –12 dB (right)
4 tom · 5 conga · 6 cowbell	Accented hits play at full volume **and higher in pitch**	How many semitones the accent goes up (0–12)

Before 3.4 the knob did nothing on channels 1, 2, 7 and 8, and on the hi-hat it swapped the accent for a fixed 808 sound. To have no accents at all, set the number of accents to 0 (shift + encoder).

Presets

80 slots: 48 factory ones – 16 green, 16 hip hop in orange and 16 latin in yellow – and **32 of your own**, stored in internal memory.

When loading and saving, **used** slots are lit and **empty** ones are dark (blue and purple user banks). While saving, shift + swEnc empties the slot.

A preset stores, for the 8 channels: pattern (K, rotation, accents), pitch, velocity, pan, linear mode, double tempo and probability; plus the sequence length and the automation.

Every preset stores its own tempo, but only imposes it if you load it while the machine is **stopped**: loading a preset then is choosing the tempo you are about to start with. While playing, the BPM you already have wins, so changing preset live never breaks the pulse, and stop/play doesn't move it either. Factory presets carry no tempo of their own and respect yours.

Not stored in the preset, because they are global: the kit, the effect and the mute state.

Backing up presets to your computer — new in 3.4

The *Backup presets* page of the webflasher can read every preset you saved, download them as a `.txt` file and **load them back into CLAVE**, each one into the slot it came from. Slots that are not in the file are left alone. The machine must be stopped. Older backup files (firmware 3.0) can't be loaded back: they don't contain the whole preset.

Drum kits

There are **9 kits**: 5 factory ones (green on the ring) and 4 user ones (purple, only if they have content).

shift + preset	Enter and leave kit selection mode
encoder	Browse the available kits (ring)
swEnc	Load the highlighted kit and exit

Uploading your own kits: from the *Upload Sound Kits* page of the webflasher you can preview 8 samples (one per channel) and save them to a user slot (1–4). They stay in the machine when you turn it off, and the chosen kit is remembered at power-up.

How to prepare your samples

- Mono WAV, 44.1 kHz, 16 or 24 bit. Whatever you upload ends up mono, 44.1 kHz, 16 bit.
- Up to **3 s per sound** and about **14 s in total** for the 8 sounds of the kit. Shorter is better.
- No silence at the start, so the hit doesn't arrive late.
- Normalize close to 0 dB: the page doesn't change the volume.
- A short fade-out at the end is **added by the machine** (new in 3.4), so a sample that ends abruptly or gets cut at 3 s won't click.
- The output is stereo: each channel has its own pan.

Effects — 14 modes

Cycle through them with mute + swEnc + encoder (one LED lights up in the colour of the effect) and set the amount (wet) with mute + encoder. Only one effect is active at a time and it is global: it is not saved in the preset.

The first 8 combine **delay** + **reverb** (magenta LED):

Room	reverb only (short)
Hall	reverb only (long)
Delay 1/8	delay only
Delay 1/4	delay only
Dub	dark delay + reverb
Slap+Room	slap + short reverb
1/4+Hall	delay + long reverb
Ambient	delay + wide reverb

The other 6 are processing effects, each with its own LED colour:

orange	SP404	Warm saturation / compression
red	Distortion	Aggressive clipping
cyan	Bitcrush	Fewer bits and lower sample rate (lo-fi)
green	Comb	Resonant comb filter, tuned to the tempo
blue	Short flanger	Narrow sweep, synced to the tempo
purple	Fast flanger	Wide, fast sweep, synced

Reverb and delay amount – new in 3.4. The bar is no longer a straight mix amount: a full bar is about 25% of what it used to be, and it rises gradually (1 click ≈ 0.2%, half bar ≈ 16%, full bar 65%). This makes the small settings usable on a drum machine, where a little goes a long way. The processing effects (SP404, distortion, bitcrush, comb, flangers) are unchanged.

The processing effects work on the signal coming in: if the overall volume is low, turn the master knob up to hear them properly.

Resonant LP filter: mute + knobs, affects the whole mix, before the effects. **Master compressor:** always on internally.

Automation

Records 2 bars of pitch (per channel) and of the filter, looping.

shift + play short	Start recording (the channel LED blinks)
play	Leave recording – the automation keeps playing
play again	Stop
shift + play 2 s	Clear

While you record one channel, the automation of the others keeps playing. The filter is only recorded if you move it. Probability is not automated: it is a fixed value per channel.

Connections and sync

Audio	1 × 3.5 mm stereo jack (output)
Clock sync	2 jacks: in and out
USB-C	2 ports: power, and data / MIDI
Power	5 V over USB-C

> ⚠ **Only connect sync in with CLAVE powered on**, and if the clock comes from a passive multiple use a buffered one (or a dedicated output on your clock): CLAVE's input can load the signal and disturb the other modules.

Clock input – new in 3.4. CLAVE works out the resolution on its own: 1, 2, 4 or 24 pulses per quarter note (PPQN). For 1, 2 and 4 PPQN it picks the one that puts the tempo between 80 and 160 BPM. If your tempo is outside that range (170 BPM would read as 85, for example), **tap TEMPO in quarter notes** with the clock connected: CLAVE picks the resolution that matches your tap and remembers it. It follows the tempo even while stopped, and when you press **play** it starts on the next pulse, in time. If the master changes BPM, CLAVE follows within one or two pulses; the more PPQN your clock sends, the tighter the tracking.

Clock output: 2 PPQN (one pulse every two steps), 50% square wave, steady tempo. Low while stopped; 3.3 V logic. Two CLAVEs chain directly: sync out of one into sync in of the other.

CLAVE also follows external **MIDI clock** over USB (24 PPQN); while USB clock is present the jack input is ignored.

While connected to the computer — new in 3.4

When one of the webflasher pages is connected to CLAVE and the machine is stopped, the ring breathes slowly in **orange**: that means "connected". While presets or samples are loading, the ring fills up like a **progress bar**, and a **green flash** means it finished. Press play and the ring goes back to showing the sequence, so you can try a kit while still connected.

Power-up combos

Hold the buttons **while you plug in the USB** (or while you reset). The light of the mode comes on as soon as the machine sees the combination, even if you are still holding it. To leave and carry on booting, **click the encoder** (swEnc); if you don't touch anything it leaves on its own after 30 s. Every setting is stored: it survives power-off.

shift + play	Download mode, to install firmware from the browser (this one enters on its own)
swEnc	Reverse the encoder direction (release it, then click again to carry on)
shift + tempo	LED brightness: turn the encoder and leave with a click
preset + play	Colour-blind palette
mute + play	Left-handed mode
play + tempo	Start-up preset: random or the last one used

Brightness. The ring, the indicators and the pads dim together. Useful for playing in the dark or for recording video. **New in 3.4:** the maximum is 50%, because the LEDs are what draws the most current and above that the touch sensor can lose power. Colours stay readable all the way down to the minimum.

Colour-blind mode. Swaps the colours of the eight channels for a palette that can be told apart with any kind of colour blindness, and keeps the two that could still be confused as far apart as possible on the ring. An accent stops being a complementary colour and becomes the same colour, brighter.

Left-handed mode. Bars are drawn from the top towards the right. By default they start at the bottom and climb up the left side, because your right hand covers the right side of the ring when you turn the encoder.

Buttons and LEDs

play	play / stop
preset	load / save
tempo	tap tempo
mute	mute and effects
shift	modifier
swEnc	encoder switch

16-LED ring: the steps of the active channel, in its colour. Accents use a different colour and empty steps stay dark. The white playback cursor only appears while playing; stopped, you see the real state of every step. In linear mode a dot also marks where the encoder cursor is.

The **bars** – effect, filter, probability, tempo, accents, length – start at the bottom and climb up the left side, so your hand doesn't cover what you are adjusting. The **mute view** uses that same left half: one LED per channel, lit if it sounds.

The 4 indicators: original pitch · sequence mode of the channel · reserved · automation (blinks while recording, steady while playing back; in the preset and kit views it shows the bank).

The 8 pad LEDs: in the mute view they show which channel sounds. Outside it, they mark the channel you are editing.

Other changes in 3.4

- **Pitch knob:** it no longer drifts on its own at the edge of the snap zone. The white pitch LED and the tuning stay put, and leaving the zone now moves by one semitone instead of two.
- **Filter and effects:** the master filter now runs on the mix before the effects, so filter + reverb no longer breaks up.
- **Kick and toms:** no more click when a sound retriggers while its tail is still ringing.

Updating the firmware

Connect the machine over USB and open the webflasher in desktop **Chrome**, **Edge** or **Brave** (it does not work in Firefox or Safari). Kits and presets are uploaded from there too.

https://clave.oficinadesonido.org

To put the machine in download mode, hold **shift** + **play** while you plug the USB cable in.

Keep your presets and kits. Before installing, the browser asks whether to erase the device and shows an **Erase device** checkbox. Leave it **unchecked**: the installer then writes only the firmware and your presets and saved kits stay in place. If you tick it, everything is wiped.

> The version installed by a given page is shown next to the Install button. This manual describes firmware 3.4.



Firmware, sound kits and preset backups:
oficinadesonido.github.io/odesEucli-flasher/mini/

Desktop Chrome, Edge or Brave. Hold **shift + play** while you plug in the USB cable to enter download mode. Installing firmware keeps your presets and kits.

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